

Formal Appendix: State–Collapse Theorem

Jinu Jang

v1.1 – June 27, 2025

0. Notation

- **Set** – category of sets and functions.
- **Vect $_{\mathbb{R}}$** – category of real vector spaces and linear maps.
- S – fixed non-empty *state set*.
- $\mathbb{R}^X$ – free vector space on X with basis $\{\delta_x\}_{x \in X}$.
- $\otimes$ – algebraic tensor product over $\mathbb{R}$.

1. The state monad T_S

$$T_S X := S \Rightarrow (S \times X).$$

Unit.

$$\eta_X(x)(s) := (s, x).$$

Multiplication.

$$\mu_X : T_S T_S X \longrightarrow T_S X, \quad \mu_X(\Phi)(s) := \Phi(s)(\pi_1 \Phi(s)).$$

Monad-algebra law.

$$\mu_X \circ T_S \mu_X = \mu_X \circ \mu_{T_S X}. \quad ()$$

2. Free T_S -algebras and Alg_{T_S}

A T_S -algebra is a pair (A, α) with $\alpha : T_S A \rightarrow A$ obeying the usual axioms. The free algebra on a set X is

$$(T_S X, \mu_X),$$

and Alg_{T_S} has morphisms $f : (A, \alpha) \rightarrow (B, \beta)$ satisfying $\beta \circ T_S f = f \circ \alpha$.

3. The vector-space functor $\mathcal{F}$

$$\boxed{\mathcal{F} : \mathbf{Set} \longrightarrow \mathbf{Vect}_{\mathbb{R}}, \quad \mathcal{F}(X) = \mathbb{R}^S \otimes \mathbb{R}^X, \quad \mathcal{F}(f) = \text{id}_{\mathbb{R}^S} \otimes \mathbb{R}^f.}$$

4. Split idempotent pair P, i

$$\begin{aligned} P_{S,X} : (\mathbb{R}^S \otimes \mathbb{R}^S) \otimes \mathbb{R}^X &\rightarrow \mathbb{R}^S \otimes \mathbb{R}^X, & P(e_{s_1} \otimes e_{s_2} \otimes \delta_x) &= e_{s_2} \otimes \delta_x, \\ i_{S,X} : \mathbb{R}^S \otimes \mathbb{R}^X &\rightarrow (\mathbb{R}^S \otimes \mathbb{R}^S) \otimes \mathbb{R}^X, & i(e_s \otimes \delta_x) &= (e_s \otimes e_s) \otimes \delta_x. \end{aligned}$$

Linearity. Both maps extend $\mathbb{R}$ -linearly from the basis.

$$P \circ i = \text{id}_{\mathbb{R}^S \otimes \mathbb{R}^X}, \quad e_{S,X} := i \circ P, \quad e_{S,X}^2 = e_{S,X}.$$

Naturality. For every function $f : X \rightarrow Y$,

$$P_{S,Y} \circ (\text{id} \otimes \mathbb{R}^f) = (\text{id} \otimes \mathbb{R}^f) \circ P_{S,X}, \quad i, e \text{ likewise.}$$

5. Reshuffle isomorphism θ

$$\theta_X : \mathcal{F}(T_S X) \xrightarrow{\cong} (\mathbb{R}^S \otimes \mathbb{R}^S) \otimes \mathbb{R}^X, \quad \theta_X(e_{s_1} \otimes \delta_{\langle s_2, x \rangle}) = e_{s_1} \otimes e_{s_2} \otimes \delta_x.$$

Inverse:

$$\theta_X^{-1}(e_{s_1} \otimes e_{s_2} \otimes \delta_x) = e_{s_1} \otimes \delta_{\langle s_2, x \rangle}.$$

6. $\mathcal{F}(\mu_X)$ explicitly

$$\boxed{\mathcal{F}(\mu_X) = \theta_X^{-1} \circ P_{S,X} \circ \theta_{T_S X}.$$

Thus $\mathcal{F}(\mu_X)$ drops the first state coordinate.

7. Collapse Theorem

Theorem 1 (State–Collapse). *For every $X \in \mathbf{Set}$,*

$$e_{S,X} = i_{S,X} \circ \mathcal{F}(\mu_X)$$

is a split idempotent in $\mathbf{Vect}_{\mathbb{R}}$, i.e. $e_{S,X}^2 = e_{S,X}$.

Proof. Using naturality and $P \circ i = \text{id}$:

$$e_{S,X}^2 = i \circ \mathcal{F}(\mu) \circ i \circ \mathcal{F}(\mu) = i \circ (P \circ i) \circ \mathcal{F}(\mu) = i \circ \mathcal{F}(\mu) = e_{S,X}.$$

□

8. Lean identifiers (reference)

Symbol	Lean name
$P_{S,X}$	state_drop
$i_{S,X}$	state_dup
$e_{S,X}$	state_idem
$P \circ i = \text{id}$	collapse_split
$e^2 = e$	collapse_idem
Collapse Theorem	collapse_theorem